P1

Lista - Limites - Matemática Elementar

34. Calcule os limites a seguir, caso existam

1a. $\lim\limits_{x \to -2} \dfrac{x-5}{4x+3} = \dfrac{-2-5}{4(-2)+3} = \dfrac{-7}{-8+3} = \dfrac{-7}{-5} = 1{,}4$

1b. $\lim\limits_{x \to -2} (3x^3 - 2x + 7) = 3(-2)^3 - 2(-2) + 7$

$= 3 . -8 + 4 + 7$

$= -24 + 11$

$= -13$

1c. $\lim\limits_{x \to -2} \dfrac{x+2}{x^2-4}$ $= a = 1 \quad b = 0 \quad c = -4$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 . 1 . -4$

$\Delta = 0 + 16$

$\Delta = 16$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-0 \pm \sqrt{16}}{2.1} = \dfrac{-0 \pm 4}{2}$

$x' = \dfrac{-0+4}{2} = \dfrac{4}{2} = 2$

$x'' = \dfrac{-0-4}{2} = \dfrac{-4}{2} = -2$

$x^2 - 4 = (x-2)(x-(-2))$ $= \lim\limits_{x \to -2} \dfrac{\cancel{x+2}}{(x-2)\cancel{(x+2)}} = (x-2) = -2-2 = -4$

$= (x-2)(x+2)$

1d. $\lim\limits_{x \to 2} \dfrac{x^2-4}{x^2-x-2}$ $= a = 1 \quad b = -1 \quad c = -2$

$\Delta = b^2 - 4ac$

$\Delta = -1^2 - 4 . 1 . -2$

$\Delta = 1 + 8$

$\Delta = 9$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-(-1) \pm \sqrt{9}}{2 . 1} = \dfrac{1 \pm 3}{2}$

$x' = \dfrac{1+3}{2} = \dfrac{4}{2} = 2$

$x'' = \dfrac{1-3}{2} = \dfrac{-2}{-2} = -1$

$x^2 - x - 2 = (x-2)(x-(-1))$

$= (x-2)(x+1)$

$\lim_{x \to 2} = \dfrac{x^2-4}{(x-2)(x+1)}$

$x^2 - 4$

$a = 1 \quad b = 0 \quad c = -4$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1(-4)$

$\Delta = 0 + 16$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-0 \pm \sqrt{16}}{2 \cdot 1} = \dfrac{-0 \pm 4}{2}$

$x' = \dfrac{0+4}{2} = \dfrac{4}{2} = 2$

$x'' = \dfrac{-0-4}{2} = \dfrac{-4}{2} = -2$

$x^2 - 4 = (x-2)(x-(-2))$

$(x-2)(x+2)$

$\lim_{x \to 2} = \dfrac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}(x+1)} = \dfrac{2+2}{2+1} = \dfrac{4}{3}$

12. $\lim_{x \to 3} \dfrac{x^2 - 25}{x - 5}$

$x^2 - 25$

$a = 1 \quad b = 0 \quad c = -25$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1(-25)$

$\Delta = 0 + 100$

$\Delta = 100$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-0 \pm \sqrt{100}}{2 \cdot 1} = \dfrac{-0 \pm 10}{2}$

$x' = \dfrac{-0+10}{2} = \dfrac{10}{2} = 5$

$x'' = \dfrac{-0-10}{2} = \dfrac{-10}{2} = -5$

$x^2 - 25 = (x-5)(x-(-5))$

$= (x-5)(x+5)$

$\lim_{x \to 5} \dfrac{\cancel{(x-5)}(x+5)}{\cancel{(x-5)}} = 5+5 = 10$

f) $\lim_{x \to -2} \frac{x^2-2x-8}{x^2+5x+6}$

x^2-2x-8

$a=1 \quad b=-2 \quad c=8$

$\Delta = b^2-4ac$

$\Delta = -2^2-4.1.8$

$\Delta = 4+32$

$\Delta = 36$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-2) \pm \sqrt{36}}{2.1} = \frac{2 \pm 6}{2}$

$x' = \frac{2+6}{2} = \frac{8}{2} = 4$

$x'' = \frac{2-6}{2} = \frac{-4}{2} = -2$

x^2+5x+6

$a=1 \quad b=5 \quad c=6$

$\Delta = b^2-4ac$

$\Delta = 5^2-4.1.6$

$\Delta = 25-24$

$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-5 \pm \sqrt{1}}{2.1} = \frac{-5 \pm 1}{2}$

$x' = \frac{-5+1}{2} = \frac{-4}{2} = -2$

$x'' = \frac{-5-1}{2} = \frac{-6}{2} = -3$

$x^2-2x-8 = (x-4)(x-(-2))$
$= (x-4)(x+2)$

$x^2+5x+6 = (x-(-2))(x-(-3))$
$= (x+2)(x+3)$

$\lim_{x \to -2} \frac{(x-4)\cancel{(x+2)}}{\cancel{(x+2)}(x+3)} = \lim_{x \to -2} \frac{-2-4}{-2+3} = \frac{-6}{1} = -6$

g) $\lim_{x \to 2} \frac{x^2+3x-10}{x^2+x-6}$

x^2+x-6

$a=1 \quad b=1 \quad c=-6$

$\Delta = b^2-4ac$

$\Delta = 1^2-4.1-6$

$\Delta = 1+24$

$\Delta = 25$

$x^2+3x-10$

$a=1 \quad b=3 \quad c=-10$

$\Delta = b^2-4ac$

$\Delta = 3^2-4.1-10$

$\Delta = 9+49$

$\Delta = 49$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-1 \pm \sqrt{25}}{2 \cdot 1} = \frac{-1 \pm 5}{2}$

$x' = \frac{-1+5}{2} = \frac{4}{2} = 2$

$x'' = \frac{-1-5}{2} = \frac{-6}{2} = -3$

$x^2 + x - 6 = (x-2)(x-(-3))$

$= (x-2)(x+3)$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-3 \pm \sqrt{49}}{2 \cdot 1} = \frac{-3 \pm 7}{2}$

$x' = \frac{-3+7}{2} = \frac{4}{2} = 2$

$x'' = \frac{-3-7}{2} = \frac{-10}{2} = -5$

$x^2 + 3x - 10 = (x-2)(x-(-5))$

$= (x-2)(x+5)$

$\lim_{x \to 2} \frac{\cancel{(x-2)}(x+5)}{\cancel{(x-2)}(x+3)} = \frac{(2+5)}{(2+3)} = \frac{7}{5} = 1,4$

b. $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 12x + 20}$

$x^2 - 5x + 6$

$a = 1 \quad b = -5 \quad c = 6$

$\Delta = b^2 - 4ac$

$\Delta = -5^2 - 4.1.6$

$\Delta = 25 - 24$

$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-5) \pm \sqrt{1}}{2 \cdot 1} = \frac{5 \pm 1}{2}$

$x' = \frac{5+1}{2} = \frac{6}{2} = 3$

$x'' = \frac{5-1}{2} = \frac{4}{2} = 2$

$x^2 - 5x + 6 = (x-3)(x-2)$

$x^2 - 12x + 20$

$a = 1 \quad b = -12 \quad c = 20$

$\Delta = b^2 - 4ac$

$\Delta = -12^2 - 4.1.20$

$\Delta = 144 - 80$

$\Delta = 64$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-12) \pm \sqrt{64}}{2 \cdot 1} = \frac{12 \pm 8}{2}$

$x' = \frac{12+8}{2} = \frac{20}{2} = 10$

$x'' = \frac{12-8}{2} = \frac{4}{2} = 2$

$x^2 - 12x + 20 = (x-10)(x-2)$

$$\lim_{x \to 2} \frac{(x-3)\cancel{(x-2)}}{(x-10)\cancel{(x-2)}} = \frac{(x-3)}{(x-10)} = \frac{2-3}{2-10} = \frac{-1}{-8} = \frac{1}{8}$$

u. $$\lim_{x \to -2} \frac{x^3+4x^2+4x}{(x+2)(x-3)} = \frac{x\cancel{(x+2)}(x+2)}{\cancel{(x+2)}(x-3)} = \frac{-2(-2+2)}{-2-3} = \frac{-20}{-5} = \frac{0}{-5}$$

$x(x^2+4x+4)=0$

x^2+4x+4

$\Delta = b^2-4ac$

$\Delta = 4^2-4.1.4$

$\Delta = 16-16$

$\Delta = 0$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm \sqrt{0}}{2.1} = \frac{-4 \pm 0}{2} = -2$$

$x^2+4x-4 = (x-(-2))(x-(-2))$

$= (x+2)(x+2)$

y. $$\lim_{t \to 0} \frac{(4+t)^2-16}{t} = 4^2+2.4t+t^2 = \lim_{t \to 0} \frac{\cancel{16}+8t+t^2-\cancel{16}}{t} = \lim_{t \to 0} \frac{8t+t^2}{t}$$

$= 16+8t+t^2$

$8t+t^2$

$a=1 \quad b=8 \quad c=0$

$\Delta = b^2-4ac$

$\Delta = 8^2-4.1.0$

$\Delta = 64-0$

$\Delta = 64$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-8 \pm \sqrt{64}}{2.1} = \frac{-8 \pm 8}{2}$$

$$x' = \frac{-8+8}{2} = \frac{0}{2} = 0$$

$$x'' = \frac{-8-8}{2} = \frac{-16}{2} = -8$$

$8t+t^2 = (x-0)(x-(-8))$

$(x-0)(x+8)$

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$$\lim_{x\to 0} \frac{(x-0)(x+8)}{x} = \frac{(0-0)(0+8)}{0} = 8$$

h. $$\lim_{x\to 16} \frac{x-16}{\sqrt{x}-4} = \frac{x-16}{\sqrt{x}-4}\left(\frac{\sqrt{x}+4}{\sqrt{x}+4}\right) = \frac{(x-16)(\sqrt{x}+4)}{x+4\sqrt{x}-4\sqrt{x}-16} = \frac{\cancel{(x-16)}(\sqrt{x}+4)}{\cancel{x-16}} = (\sqrt{x}+4)$$

$$= \sqrt{16}+4 = 4+4 = 8$$

i. $$\lim_{x\to 0} \frac{\sqrt{16+x}-4}{x} = \frac{\sqrt{16+x}-4}{x}\cdot\frac{\sqrt{16+x}+4}{\sqrt{16+x}+4} = \frac{16+x+4\sqrt{16x}-4\sqrt{16x}-16}{x(\sqrt{16+x}+4)}$$

$$= \frac{\cancel{16}+x-\cancel{16}}{x\sqrt{16+x}+4} = \frac{\cancel{x}}{\cancel{x}\sqrt{16+x}+4} = \frac{1}{\sqrt{16+0}+4} = \frac{1}{4+4} = \frac{1}{8}$$

35. Calcule os limites a seguir, caso existam.

a. $$\lim_{x\to 4} \frac{5}{x-4} = \lim_{x\to 4} \frac{5}{(4-4)^1} = \frac{5}{0} = \nexists$$

b. $$\lim_{x\to -\frac{5}{2}} \frac{8}{(2x+5)^3} = \lim_{x\to -\frac{5}{2}} \frac{8}{\left(\frac{2\cdot -5}{2}+5\right)^3} = \frac{8}{(-5+5)^3} = \frac{8}{0^3} = \frac{8}{0} = \nexists$$

c. $$\lim_{x\to -1} \frac{-1}{(x+1)^2} = \frac{-1}{(-1+1)^2} = \frac{-1}{0^2} = \frac{-1}{0} = -\infty$$

d. $$\lim_{x\to \frac{9}{2}} \frac{2x}{(2x-9)^2} = \frac{\frac{2\cdot 9}{2}}{\frac{2\cdot 9}{2}-9} = \frac{\frac{18}{2}}{\frac{18}{2}-9} = \frac{9}{\frac{18-18}{2}} = \frac{9}{0} = +\infty$$

e. $\lim\limits_{x\to 3} \dfrac{1}{x(x-3)^4} = \dfrac{1}{3(3-3)^4} = \dfrac{1}{(9-9)^4} = \dfrac{1}{0^4} = \dfrac{1}{0} = +\infty$

f. $\lim\limits_{x\to 2} \dfrac{1}{|x-2|} = \dfrac{1}{2-2} = \dfrac{1}{0} = +\infty$

g. $\lim\limits_{x\to\infty} \dfrac{5x^2-3x+1}{2x^2+4x-1} = \dfrac{5x^2}{2x^2} = \dfrac{5}{2}$

h. $\lim\limits_{x\to-\infty} \dfrac{-1x+4}{2+3x} = \dfrac{-1x}{3x} = \dfrac{1}{3}$

i. $\lim\limits_{x\to-\infty} \dfrac{2x^2-3}{4x^3+5x} = \dfrac{2x^2}{4x^3} = \dfrac{2}{4x} = \dfrac{2}{-\infty} = 0$

j. $\lim\limits_{x\to\infty} \dfrac{2x^2-x+3}{x^3+1} = \dfrac{2x^2}{x^3} = \dfrac{2}{x} = \dfrac{2}{\infty} = 0$

NT

k. $\lim\limits_{x\to\infty} \dfrac{-x^3+2x}{2x^2-3} = \dfrac{-x^3}{2x^2} = \dfrac{-x}{2} = \dfrac{-\infty}{2} = -\infty$

NT

l. $\lim\limits_{x\to-\infty} \dfrac{2-x^2}{x+3} = \dfrac{-x^2}{x} = -x = +\infty$

36. Seja $f(x) = \begin{cases} x-1, & x \leq 2 \\ 2x-3, & x > 2 \end{cases}$ Esboce seu gráfico e:

menor ou igual

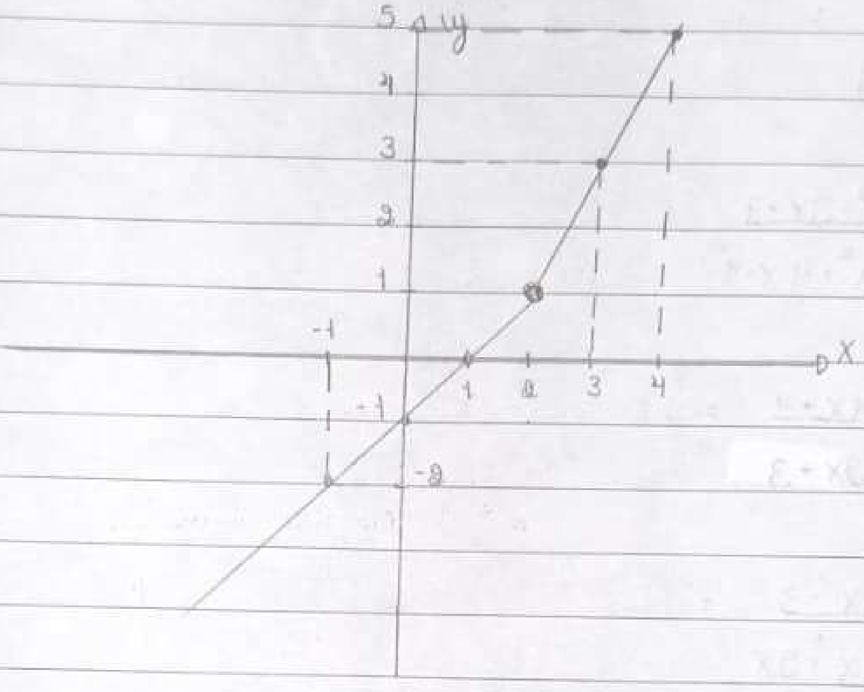
maior

a. Determine

- $\lim_{x \to 2^+} f(x)$ $\lim_{x \to 2^+} f(x) = 2x - 3 \rightarrow$ Maior que 2 $= 2.2 - 3 = 1$

- $\lim_{x \to 2^-} f(x)$ $\lim_{x \to 2^-} f(x) = x - 1 \rightarrow$ Menor ou igual a 2 $= 2 - 1 = 1$

x	2x-3
3	3
4	5
5	1
2	1



x	x-1
1	0
0	-1
-1	-2
2	1

Imagem = $\mathbb{R}$

- $\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} f(x) = 2x - 3 = 2x = +\infty$

- $\lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} f(x) = x - 1 = x = -\infty$

2. A função $f(x)$ é contínua em $x = 2$? Justifique.

Sim, pois $\lim_{x \to 2^+} = \lim_{x \to 2^-}$

3ª. Seja $f(x) = \begin{cases} x^2 - 1 & \text{se } x \geq 0 \\ -x + 2 & \text{se } x < 0 \end{cases}$. Esboce o gráfico.

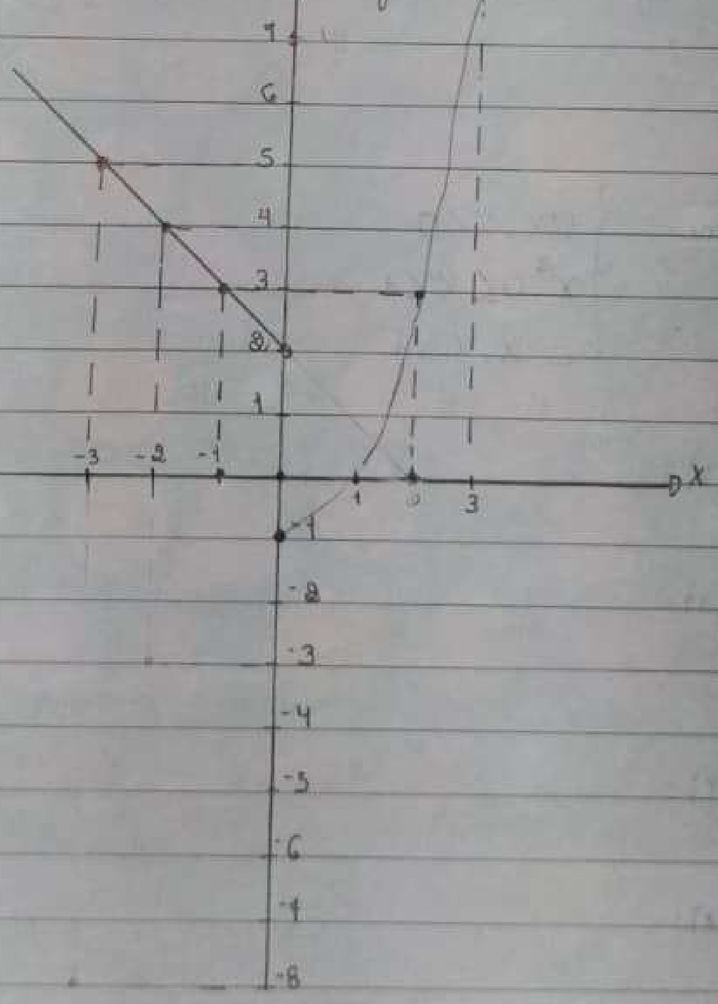
a. Determine

* $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} f(x) = x^2 - 1$ → Maior ou igual a $0 = 0^2 - 1 = 0 - 1 = -1$

* $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} f(x) = -x + 2$ → Menor do que $0 = -0 + 2 = 2$

$f(x)$	x^2-1
-1	0
2	3
3	8
0	-1

$f(x)$	$-x+2$
-1	3
-2	4
-3	5
0	2



(/ /)

* $\lim_{x \to +\infty} f(x)$ $\lim_{x \to +\infty} f(x) = x^2 + 1 = x^2 = +\infty$

* $\lim_{x \to -\infty} f(x)$ $\lim_{x \to -\infty} f(x) = -x + 2 = -x = +\infty$

↳ sempre que for $-\infty - = +\infty$

b. A função f(x) é contínua em x = 0? Justifique

Não é contínua pois para o $\lim_{x \to 0+} \neq \lim_{x \to 0-}$

38. Seja $f(x) = \begin{cases} 1/x, & x < 0 \\ x^2, & 0 \leq x < 1 \\ 2 - x, & x > 1 \end{cases}$

menor que 0

0 maior ou igual X ou X menor que 1

X maior que 1

a. $\lim_{x \to 0+} f(x) = x^2 \cdot \lim_{x \to 0+} f(x) = 0^2 = 0$

* b. $\lim_{x \to 0-} f(x) = \frac{1}{x} = \lim_{x \to 0-} f(x) = \frac{1}{0-} = -\infty$

c. $\lim_{x \to 0} f(x) = x^2 = \lim_{x \to 0} 0^2 = 0$

d. $\lim_{x \to 1} f(x) = \cancel{\exists}$

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39 Determine, se existirem, os pontos onde as seguintes funções não são contínuas.

a. $f(x) = \dfrac{x}{(x-3)(x+7)} = f(x) = \dfrac{x}{x^2+7x-3x-21} = \dfrac{x}{(x^2+4x-21)} \neq 0$

$x^2+4x-21$

$a=1 \quad b=4 \quad c=-21$

$\Delta = b^2-4ac$

$\Delta = 4^2-4.1.-21$

$\Delta = 100$

$x = \dfrac{-b\pm\sqrt{\Delta}}{2a} = \dfrac{-4\pm\sqrt{100}}{2.1} = \dfrac{-4\pm 10}{2}$

$\to x' = \dfrac{-4+10}{2} = \dfrac{6}{2} = 3$

$\to x'' = \dfrac{-4-10}{2} = \dfrac{-14}{2} = -7$

Domínio = $\{x \in \mathbb{R} / x \neq 3$ e $x \neq -7\}$

b. $f(x) = \dfrac{x^2+3x-1}{x^2-6x+8}$

x^2-6x+8

$a=1 \quad b=-6 \quad c=8$

$\Delta = b^2-4ac$

$\Delta = (-6)^2-4.1.8$

$\Delta = 36-32$

$\Delta = 4$

$x = \dfrac{-b\pm\sqrt{\Delta}}{2a} = \dfrac{-(-6)\pm\sqrt{4}}{2.1} = \dfrac{6\pm 2}{2}$

$\to x' = \dfrac{6+2}{2} = \dfrac{8}{2} = 4$

$\to x'' = \dfrac{6-2}{2} = \dfrac{4}{2} = 2$

Domínio = $\{x \in \mathbb{R} / x \neq 4$ e $x \neq 2\}$

40. Determine, se as funções a seguir são contínuas ou não nos valores de a dados.

a. $f(x) \begin{cases} \dfrac{x^2 - x}{x^2 - 1} & \text{se} \neq 1 \\ 1 & \text{se} \neq 1 \end{cases}$ $\quad a = 1$

$$\lim_{x \to 1} \frac{x^2 - x}{x^2 - 1} = \frac{1^2 - 1}{1^2 - 1} = \frac{0}{0} = \lim_{x \to 1} \frac{(x-0)\cancel{(x-1)}}{\cancel{(x-1)}(x+1)} = \lim_{x \to 1} \frac{(x-0)}{(x+1)} = \frac{1-0}{1+1} = \frac{1}{2} = 0{,}5$$

$x^2 - x$

$a = 1 \quad b = -1 \quad c = 0$

$\Delta = b^2 - 4ac$

$\Delta = -1^2 - 4 . 1 . 0$

$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-1) \pm 1}{2 . 1} = \frac{+1 \pm 1}{2}$

$\to x' = \frac{1 - 1}{2} = \frac{0}{2} = 0$

$\to x'' = \frac{+1 + 1}{2} = \frac{2}{2} = 1$

$x^2 - x = (x - 0)(x - 1))$

$= (x - 0)(x - 1)$

$x^2 - 1$

$a = 1 \quad b = 0 \quad c = -1$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 . 1 . (-1)$

$\Delta = 0 + 4$

$\Delta = 4$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-0 \pm \sqrt{4}}{2 . 1} = \frac{0 \pm 2}{2}$

$\to x' = \frac{0 + 2}{2} = \frac{2}{2} = 1$

$\to x'' = \frac{0 - 2}{2} = \frac{-2}{2} = -1$

$x^2 - 1 = (x - 1)(x - (-1))$

$= (x - 1)(x + 1)$

$\lim_{x \to 1} 1 = 1$

Elas não são contínua porque $\lim_{x \to 1} \frac{x^2 - x}{x^2 - 1} \neq \lim_{x \to 1} 1$

12. $f(x) = \begin{cases} \frac{x^2 - x - 12}{x+3} & \text{se } x \neq -3 \\ -5 & \text{se } x = -3 \end{cases}$

$\lim_{x \to -3} \frac{x^2 - x - 12}{x+3} = \lim_{x \to -3} \frac{9 + 3 - 12}{-3+3} = \frac{12 - 12}{-3+3} = \frac{0}{0}$

$x^2 - x - 12$

$a = 1 \quad b = -1 \quad c = -12$

$\Delta = b^2 - 4ac$

$\Delta = (-1)^2 - 4 \cdot 1 (-12)$

$\Delta = 1 + 48$

$\Delta = 49$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-1) \pm \sqrt{49}}{2 \cdot 1} = \frac{1 \pm 7}{2}$

$\to x' = \frac{1+7}{2} = \frac{8}{2} = 4$

$\to x'' = \frac{1-7}{2} = \frac{-6}{2} = -3$

$x^2 - x - 12 = (x-4)(x-(-3))$

$= (x-4)(x+3)$

$\lim_{x \to -3} \frac{(x-4)\cancel{(x+3)}}{\cancel{(x+3)}} = (x-4) = (-3-4) = -7$

$\lim_{x \to -3} -5 = -5$

Ela não são contínuas porque $\lim_{x \to -3} \frac{x^2 - x - 12}{x+3} \neq \lim_{x \to -3} -5$

1c. $f(x) = \begin{cases} 1 + x^2 & \text{se } x < 1 \\ 3 - x & \text{se } x \geq 1 \end{cases}$ $a = 1$

$\lim_{x \to 1^+} 1 + x = \lim_{x \to 1^+} 1 + 1 = 2$

$\lim_{x \to 1^-} 3 - x = \lim_{x \to 1^-} 3 - 1 = 2$

São contínuas porque $\lim_{x \to 1} 1 + x = \lim_{x \to 1} 3 - x$

11. Determine os valores de a para que as funções abaixo sejam contínuas em $\mathbb{R}$.

a. $f(x) = \begin{cases} x + 2a & \text{se } x < -1 \\ ax + 2 & \text{se } x \geq -1 \end{cases}$

$\lim_{x \to -1^+} ax + 2 = -a + 2$ $\lim_{x \to -1^-} x + 2a = -1 + 2$

$-a + 2 = -1 + 2a$

$2 + 1 = 2a + a$

$3 = 3a$

$a = 1$

b. $f(x) = \begin{cases} ax + 1 & \text{se } x \leq 3 \\ ax^2 - 1 & \text{se } x > 3 \end{cases}$

$\lim_{x \to 3^+} ax^2 - 1 = 9a - 1$ $\lim_{x \to 3^-} ax + 1 = 3a + 1$

$9a - 1 = 3a + 1$

$6a = 2$

$a = \frac{6}{2} = 3$

1c- $f(x) = \begin{cases} x^2 - a & \text{se } x < 2 \\ ax + 20 & \text{se } x \geq 2 \end{cases}$

$\lim\limits_{x \to 2^+} ax + 20 = 2a + 20$

$\lim\limits_{x \to 2^-} x^2 - a = 4 - a$

$2a + 20 = 4 - a$

$3a = 4 - a$

$3a = 4 - 20$

$a = \dfrac{-16}{3}$